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Assignment 7

Interim Submission

Due Date: June 6th, 2025.

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Problem Statement

Emergency departments in resource-constrained Caribbean settings like Mercer General Hospital operate on paper-based triage systems that generate no structured, queryable data. Without a digital foundation, locally validated AI triage tools cannot be built, and globally trained models — developed predominantly on North American and European patient populations — cannot be reliably applied. This project proposes to establish a structured digital triage dataset from the Mercer General ED as the necessary first step toward a clinically credible, regionally relevant AI-assisted triage classifier.

Summaries

Paper 1 — Tyler et al. (2024)

Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. Cureus.

This scoping review examined the state of AI and machine learning in ED triage, asking whether these tools meaningfully improve triage outcomes. The authors searched EMBASE, Ovid MEDLINE, and Web of Science, screening over 1,140 citations and applying strict inclusion criteria to peer-reviewed US-based primary research published between 2013 and 2023. Findings showed that AI algorithms have demonstrated potential for earlier diagnosis and faster intervention, but the authors flagged overconfident outputs as a patient safety concern. A key limitation was geographic scope — restricting to US journals excludes the majority of the world's ED contexts, including the Caribbean entirely.

Paper 2 — Da'Costa et al. (2025)

AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. International Journal of Medical Informatics.

This narrative review examined AI-driven triage systems across the literature from 2015 to 2024, sourced from PubMed, Scopus, IEEE Xplore, and Google Scholar. The authors found that AI systems automating patient prioritisation using real-time vital signs, medical history, and presenting symptoms show promise in addressing ED overcrowding and inconsistent human triage, particularly during peak hours and mass casualty events. Benefits identified included improved prioritisation speed and reduced variability in acuity assignment. The authors noted that most systems reviewed assume stable infrastructure — reliable power, internet connectivity, and EHR integration — conditions that are not guaranteed in resource-constrained settings.

Paper 3 — Araouchi & Adda (2024)

TriageIntelli: AI-Assisted multimodal triage system for health centres. Procedia Computer Science.

This paper proposed and evaluated TriageIntelli, a multimodal AI triage system that combines structured vital signs data with unstructured free-text clinical notes to improve triage accuracy beyond what single-modality approaches achieve. The system was designed for health centres rather than large academic hospitals, positioning it as a more accessible deployment target.

Results showed improved classification accuracy compared to vitals-only models, suggesting that incorporating clinical narrative adds a meaningful signal. However, the system was validated exclusively on datasets from well-resourced settings with complete electronic records — a precondition that a paper-based ED like Mercer General does not currently meet.

References

1. Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Lee, D. W., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2024). Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. *Cureus, 16*(5), e59906. https://doi.org/10.7759/cureus.59906
2. Da'Costa, A., Teke, J., Origbo, J. E., Osonuga, A., Egbon, E., & Olawade, D. B. (2025). AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *International Journal of Medical Informatics, 197*, 105838. https://doi.org/10.1016/j.ijmedinf.2025.105838
3. Araouchi, Z., & Adda, M. (2024). TriageIntelli: AI-Assisted multimodal triage system for health centers. *Procedia Computer Science, 251*, 430–437. https://doi.org/10.1016/j.procs.2024.11.130